

Dummy variable regression models:

* In chapter one we discussed briefly the four types of variable that are generally encountered (और इसमें) in empirical analysis.

* These are - ratio scale, interval scale, ordinal scale and nominal scale ratio scale, interval scale, ordinal scale and nominal scale.

* The types of variables that we have encountered in the preceding chapters were essentially ratio scale.

* But, this ^{should} ~~could~~ not give the impression that regression models can deal only with ratio scale variables.

* Regression model can also handle other types of variable.

* In this chapter, we consider models that may involve not only ratio scale variable but also nominal scale variables.

* Such variable are also known as "indicator variables", "categorical variables", "qualitative variables" or "dummy variables".

The nature of Dummy variables:

* In regression analysis the dependent variable is frequently influenced not only by ratio scale variables (such as income, output, prices, cost, height, temperature etc.) but also by variables that are essentially qualitative or nominal scale in nature (such as sex, race, color, religion, city, geographical region) etc.

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* For example, holding all other ^{factors} constant, female workers earn less than male workers, or non-white workers earn less than white workers.

* This pattern may result from sex or racial discrimination.

* The way to find the dummy variable is 0 and 1 process.

* If there, 1 indicates the presence of that attribute and 0 indicates the absence of that attribute.

* For an example— 1 may indicate that a person is a female and 0 may designate (नहीं) a male.

* Variables that assume such 0 and 1 values are called dummy variables.

* As a matter of fact a regression model may contain regressions that are all exclusively dummy or qualitative in nature.

* Such models are called analysis of variance (ANOVA) models. (7)

ANOVA models: To illustrate the ANOVA models, consider the following examples—

* Public School Teachers Salaries by Geographical region.

* Suppose we want to find out if the average annual salary of public school teachers differs among the three geographical regions of the country.

नतो

* The three geographical categories are —

- ① Central
- ② North and
- ③ South.

* So, the following model is —

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i$$

where —

Y_i = Average salary of public school teacher in state i .

$D_{2i} = 1$; which is the North region and —
 $= 0$; otherwise.

$D_{3i} = 1$; which is the South region and —
 $= 0$; otherwise.

Now —

$$\hat{Y}_i = 26000 - 1700 D_{2i} - 3000 D_{3i}$$

$$se = (1128) \quad (1435) \quad (1499)$$

$$t = (23.18) \quad (-1.208) \quad (-2.18)$$

$$P\text{-value} = (0.0000) \quad (0.2330) \quad (0.0349)$$

$$R^2 = 0.0901.$$

* Here, $\hat{\beta}_1 = 26000$ is the mean annual consumption income¹ in the central region. ✓

* If, $E(Y_i | D_{2i}=1, D_{3i}=0)$ then —

$\hat{\beta}_2 = \hat{\beta}_1 - \beta_2 = 26000 - 1700 = 24300$; which is the annual income in the North region.

* If, $E(Y_i | D_{2i}=0, D_{3i}=1)$ then —

$\hat{\beta}_3 = \hat{\beta}_1 - \beta_3 = 26000 - 3000 = 23000$; which is the annual income in the South region.

Researcher's
Principle

⑦ If a qualitative variable has more than one category, as in our illustrative example, the choice of the benchmark category is strictly up to the researcher.

In our illustrative example, we could have chosen the south as the benchmark category, ~~is strictly~~ in that case ~~then~~ the regression results will change. Because all comparisons are made in relation to the south. This will not change the overall conclusion.

(vi) Here, we warned about the dummy variable trap. There is a way to overcome this trap by introducing as many dummy variables as the number of categories of that variable. ~~here~~

Here we do not introduce the intercept in such a model.

Here, we drop a intercept term and then get the following model—

$$Y_i = \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + U_i$$

Here, we do not dummy variable trap because there is no intercept term.

Here—

β_1 = mean salary of teachers in the central.

β_2 = mean salary of teachers in the North and

β_3 = mean salary of teachers in the south.

Say for example—

$$Y_i = 26000 + 24000D_{2i} + 22000D_{3i}$$

$$SE = (11.57) \quad (887.9130) \quad (926.8345)$$

$$t = (23.1720) \quad (27.5171) \quad (23.1987)$$

at 1% level.

where indicates that the p-values of these t ratios are very small.

Here, we can see, the dummy coefficients give directly the mean salary in the three region—central, north and south.

(vii) Here, we discuss, which is a better method of introducing a dummy variable—

(i) introduce a dummy for each category and omit (reference) the intercept term, and

(ii) include the intercept term and introduce only $(m-1)$ dummies, where m is the number of categories of the dummy variable.

Anova model with two qualitative variables:

* In the previous section we considered an ANOVA model with one qualitative variable with three categories.

* In this section we consider another ANOVA model but with two qualitative variables and bring some additional point about dummy variables.

* Say for an example: Hourly wage in relation to marital status and male and female region of residence.

$$Y_i = 8.81 + 1.10D_{2i} - 1.67D_{3i}$$

$$SE = (0.401) \quad (0.464) \quad (0.485)$$

$$P\text{-value} = (0.0000) \quad (0.0132) \quad (0.0006)$$

$$R^2 = 0.0322$$

* where,

$$Y = \text{Hourly wage}$$

$$D_2 = \text{male}$$

$$D_3 = \text{female}$$

married and 0 = otherwise
and 0 = 1 otherwise

* In this two qualitative regression and each with two categories.

* Here, the benchmark category is ~~not~~ unmarried and non-south region.

* Here, unmarried people are not in the south region and which are the ~~omitted~~ omitted category.

* Here, the mean hourly wage in the benchmark is about $\hat{\beta}_1 = 8.81$.

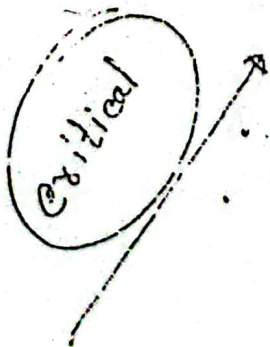
if, $E(Y_i/D_{2i}=1, D_{3i}=0)$ then,

$\hat{\beta}_2 = (\hat{\beta}_1 + \hat{\beta}_2) = 8.81 + 1.10 = \9.91 which is the mean average ^{hourly} wage of married person and who live in the south region.

if $E(Y_i/D_{2i}=0 \text{ and } D_{3i}=1)$ then—

$\hat{\beta}_3 = (\hat{\beta}_1 - \hat{\beta}_2) = 8.81 - 1.67 = \7.14 which is the average wage hourly wage of married person and who live in the south region.

* If we ~~both side~~ set 0 in both side then we find only $\hat{\beta}_1$ which is benchmark category / hourly wage for unmarried person and live in the non-south region.



Regression with a matrix of quantitative and Qualitative Regressor: The ANCOVA models =

* Generally in most economic research a regression model contains some explanatory variables that are quantitative and some that are qualitative.

Regression models containing an admixture of quantitative and qualitative variables are called analysis of covariance (ANCOVA) models.

* In a model that includes both quantitative and qualitative or dummy regressor. Thus the following model is —

$$Y_i = \beta_1 + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_i + u_i$$

where, Y_i = average annual salary of public school teachers in state (\$)

X_i = spending on public school per pupil (\$)

$D_{2i} = 1$, if the state is North.
= 0, otherwise.

$D_{3i} = 1$, if the state is South.
= 0, otherwise.

* Here, the central region is the benchmark category.

* Here, we noticed that, there are two qualitative regressors and we have a one quantitative variable X .

* Thus the numerical example —

$$\begin{aligned} \hat{Y}_i &= 13000 - 1600 D_{2i} - 1100 D_{3i} + 3.29 X_i \\ SE &= (1395.06) \quad (801.17) \quad (861.12) \quad (0.3176) \\ t &= (9.3115) \quad (-2.0889) \quad (-1.285) \quad (10.359) \end{aligned}$$

$$R^2 = 0.7266$$

P < 0

* Here a public school teacher's Salary goes up by about \$3.29, controlling for spending on education.

* mean salary of central region is —

$$\beta_1 = 13000 + 3.29x_i$$

* if $E(Y_i | D_{2i}=1; D_{3i}=0)$ then —

$$\beta_2 = (\beta_1 - \beta_3) + 3.29x_i = (13000 - 1100) + 3.29x_i = 11900 + 3.29x_i$$

which is the average salary of north region.

* if $E(Y_i | D_{2i}=0; D_{3i}=1)$ then —

$$\beta_3 = (\beta_1 - \beta_2) + 3.29x_i = (13000 - 11900) + 3.29x_i = 11100 + 3.29x_i$$

which is the average salary of south region.

* now, we show it graphically —

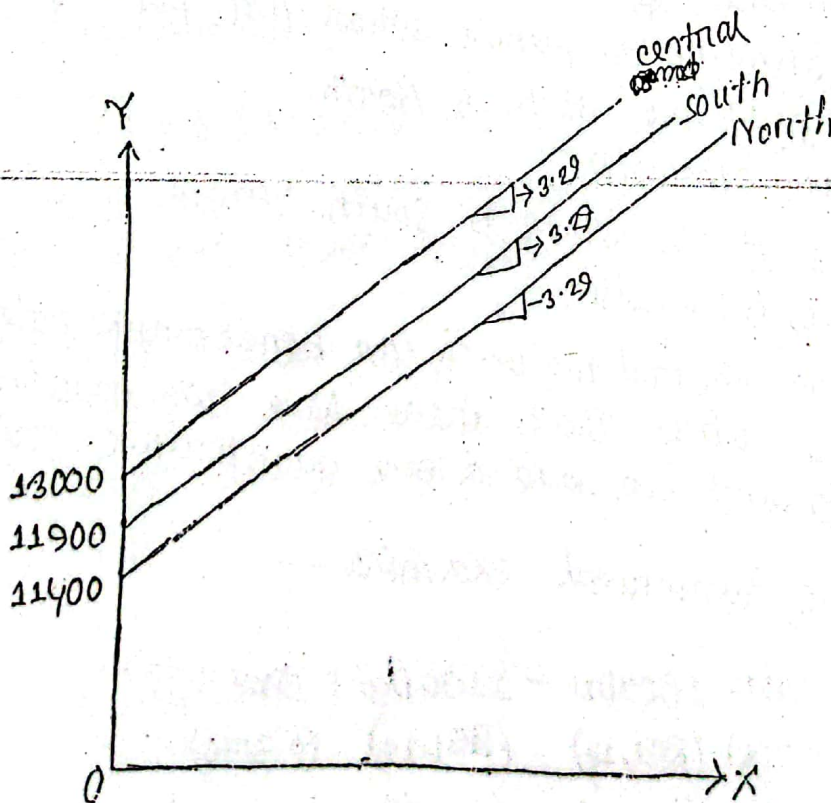


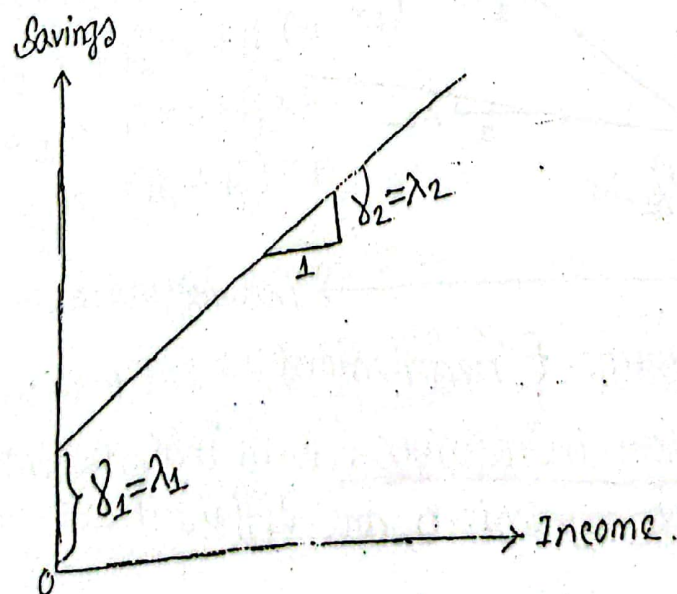
Figure Public school teacher's salary (Y) in relation to per pupil expenditure on education (X)

★ Testing structural stability using dummy variables (Chow Test) =

- * When we use a regression model involving time series data, it may happen that, there is a structural change in the relationship between the regressand Y and the regressors.
- * By structural change, we mean that, the values of the parameters of the model do not remain the same through the entire time period.
- * Sometimes the structural change may be due to external forces.
- * Chow test is the test, which tells whether there exist structural change or not.

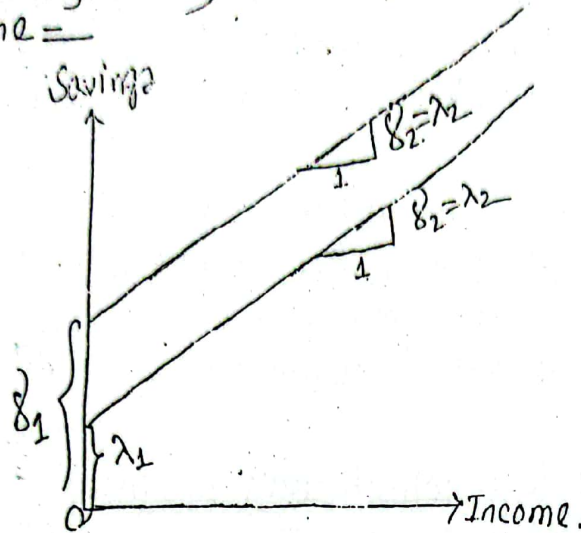
* There are four possibilities in chow test —

① Coincident regression: Both the intercept and the slope co-efficients are the same in the two regression =



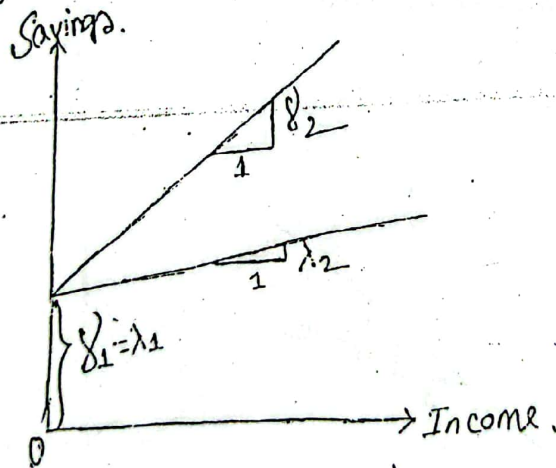
② coincident regression.

(ii) Parallel regression: only the intercepts in the two regressions are different but the slopes are the same =



ⓑ Parallel regression.

(iii) concurrent regression: The intercepts in the two regressions are the same, but the slopes are different =



ⓒ concurrent regression.

(iv) Dissimilar regression: Both the intercepts and slopes in the two regressions are different =



* The Chow-test regression function is =

$$Y_t = \alpha_1 + \alpha_2 D_t + \beta_1 X_t + \beta_2 (D_t X_t) + u_t$$

where,

Y = Savings

X = income

t = time

$D = 1$ for observation in 1970-1981

$= 0$ for otherwise (1982-1995)

$$Y_i = \alpha_1 + \alpha_2 D_i + \beta_1 X_i + \beta_2 (D_i X_i) + u_i$$

* Mean Saving function for 1970-1981

$E(Y_i | D_i = 0, X_i)$ then -

$$= \alpha_1 + \alpha_2 D_i + \beta_1 X_i + \beta_2 (D_i X_i)$$

$$= \alpha_1 + \alpha_2 \cdot 0 + \beta_1 X_i + \beta_2 (0 \cdot X_i)$$

$$= \alpha_1 + \beta_1 X_i$$

* Mean Saving function for 1982-1995

$E(Y_i | D_i = 1, X_i)$ then -

$$= \alpha_1 + \alpha_2 D_i + \beta_1 X_i + \beta_2 (D_i X_i)$$

$$= \alpha_1 + \alpha_2 \cdot 1 + \beta_1 X_i + \beta_2 (1 \cdot X_i)$$

$$= \alpha_1 + \alpha_2 + \beta_1 X_i + \beta_2 X_i$$

$$= (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) X_i$$

* now, the numerical example =

$$\hat{Y}_t = 1.0161 + 152.48 D_t + 0.0803 X_t - 0.0655 (D_t X_t)$$

$$se = (20.1648) \quad (33.0824) \quad (0.0144) \quad (0.0159)$$

$$t = (0.0504) \quad (4.6098) \quad (5.5413) \quad (-4.0963)$$

$$R^2 = 0.8819$$

$p > 0$

1970-1981
1982-1995



Mean Saving
function for 1970

* Savings income regression 1970-1981 = If $D=0$

$$\hat{Y}_t = 1.0161 + 0.0803 X_t$$

* Savings income regression 1982-1995 = If $D=1$

$$\hat{Y}_t = (1.0161 + 152.48) + (0.0803 - 0.0655) X_t$$

$$= 153.4961 + 0.0148 X_t$$

Solve the following questions:

$$\log(\text{wage}) = 0.56 - 0.324 \text{ female} + 0.087 \text{ educ} - 0.0007 \text{ female} \cdot \text{educ} + 0.034 \text{ exper} - 0.00063 \text{ exper}^2$$

$(0.32) \quad (0.16) \quad (0.036) \quad (0.0021)$

$$n=526, R^2=0.563 \quad (0.013) \quad (0.00033)$$

where, wage = '000 in a year, educ, exper = education and experience in years.

(i) control group: the category for which a dummy variable is assigned is called the control group. In this model male is a control group because if there is no assigned dummy variable.

Experimental group: The category for which a dummy variable is assigned is called the experimental variable. In this model female is a experimental group, because we are comparing a female wage per hour with the male wage.

(iii) In this model female earn less than the male 0.324 or 32.4%. Hence female income is $(0.56 - 0.324) = 0.236$ or 23.6% and male income is 0.56 or 56%. So we can say that the male earn more than the female. now $t\text{-stat} = \frac{0.324}{0.16} = 2.025$, we can say that the evidence is strong.

P.T.O

(iv) when educ is fixed—

$$\frac{\Delta \log(\text{wage})}{\Delta \text{female}} = 0.324 - 0.0067 \text{ educ.}$$

Here female earn less than male by 32.4% irrespective of education but as schooling year of female income her returns also increase but it is less than that of male. when female is fixed—

$$\frac{\Delta \log(\text{wage})}{\Delta \text{educ}} = 0.087 - 0.0067 \text{ educ}$$

Here, with education male return is 8.7% but female return is 0.67%. that indicates the female return is less than the male return. which is $(0.087 - 0.0067) = 0.0793$ or 7.93%. when educ is 16 years then,

$$\log(\text{wage}) = 0.56 - 0.324(1) + 0.087(16) - 0.0067(16)$$

$$= 1.521$$

$$\text{wage} = \text{Antilog}(1.521) = 4.58$$

⑤ The marginal effect of age will be negative at the following point—

Here,

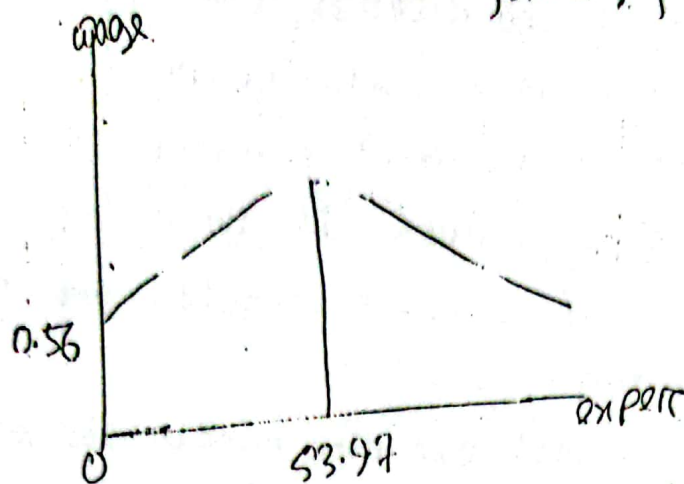
$$\text{exper} \left| \frac{\hat{\beta}_5}{\hat{\beta}_6} \right|$$

$$= \left| \frac{0.034}{0.00063} \right|$$

$$= 53.97$$

pt.0

it can be shown in the following figure—



After the 53.97 years wage rate tends to decline.
At any point above $\text{exper}^* = 53.97$ marginal effect of experience of wage will be negative.

(vi) we know—

$$\begin{aligned}\text{Adjusted } R^2 (\bar{R}^2) &= 1 - (1 - R^2) \frac{n-1}{n-k} \\ &= 1 - (1 - 0.563) \frac{526-1}{526-6} \\ &= 1 - (0.437)(1.009) \\ &= 1 - 0.441 \\ &= 0.559\end{aligned}$$

Adjusted R^2 shows goodness of fit like R^2 but it is adjusted with d.f. for excluding the effect of irrelevant variable. Here for $k > 1$, $\bar{R}^2 < R^2$ means that as R^2 is increasing as variable is increasing but it increases less than that of R^2 .

① If the wage rate at 00 (hundred unit) to 000 then the wage rate decline 10%.

Dummy Variable:

1. In econometrics we discuss four types of variable. These are ratio scale, interval scale, ordinal scale and nominal scale. These variable are also called the ratio scale.

2. These types of variable should not give the impression.

3. Regression model can also handle the other types of variable. But

4. Now, we consider a model that not only discuss the ratio scale but also discuss the nominal scale.

5. In regression analysis the dependent variable is influenced not only quantitative variable but also a qualitative variable.

6. So, these types of variable are called the dummy variable.

Say for example:

In econometrics the time series data analysis dummy variable may be used to indicate the occurrence of war or a major strike. This could be represented as a numerical value 0 or 1.

So, it is called the dummy variable.

Dummy variable trap:

1. If the dummy variable for all categories are included their sum will be equal to 1.
 2. Which is identical and perfectly correlated with the vector of ones, whose coefficient is the constant term.
 3. If the vector of ones variable are also present this will result in perfect multi co-linearity.
 4. So, that the matrix inversion the estimation algorithm would be impossible.
- So, this is referred to the dummy variable trap.

Say, for example:

Let ...

Say - 12,000 D_1

and - 10,000 D_2

D_1 / D_2

$(1 \ 1)$

wage : ... - 3000 D_1 - 5000 D_2

$(2-1) (1 \ 1)$

Dummy Variable:

In the previous chapter we discuss four types of variable these are, ratio scale, interval scale, ordinal scale and nominal scale. These types of variable were essentially ratio scale. This types of variable should not give the impression. Regression models can also handle other types of variable. But now we consider a model that not only involves the ratio scale ~~but also~~ variables but also nominal scale variables. Such variable are also known as indicator variables or dummy variables.

OR

In regression analysis the dependent variables are influenced not only by quantitative variables but also by qualitative variables.

In Statistics and econometrics particularly, in regression analysis a dummy variable is one that takes the value 0 ~~or~~ 1. ^{or} 1 to indicate the absence or presence of some category that may be expected to shift the outcome.

For example:

In econometrics time series analysis dummy variable may be used to indicate the occurrence of wars on major strikes. It could thus be thought of as a truth value represented as a numerical value 0 or 1. So, it is also called the dummy variable.

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Dummy variable trap:

1- If dummy variable for all categories are included their sum will be equal to 1 for all observations which is identical to and hence perfectly co-related with the vector of ones variable whose coefficient is the constant term. If the vector of one's variable are also present this will result in perfect multi-collinearity. So that the matrix inversion in the estimation algorithm would be impossible. This is referred to as the dummy variable trap.

For example: consider the following simple model of hourly wage determination —

$$\text{wage} = \beta_0 + \beta_1 \text{female} + \beta_2 \text{educ} + u.$$

There are just two groups, we only need two different intercepts. This means that in addition to β_0 , we need to use a one dummy variable. Using two dummy would introduce perfect co-linearity because $\text{female} + \text{male} = 1$. Including dummy variables for both genders is the simplest example of the so-called dummy variable trap, which arises when too many dummy variables describe a given number of group.

P.T.O

when female is fixed

$$\frac{\Delta \log(\hat{wage})}{\Delta educ} = 0.013 - 0.0016 \cdot female$$

with education male's return is 1.3% female return with education is less than that of male by 56%. Her return with education is $(0.013 - 0.0056) = 0.0074$ or 0.74%.

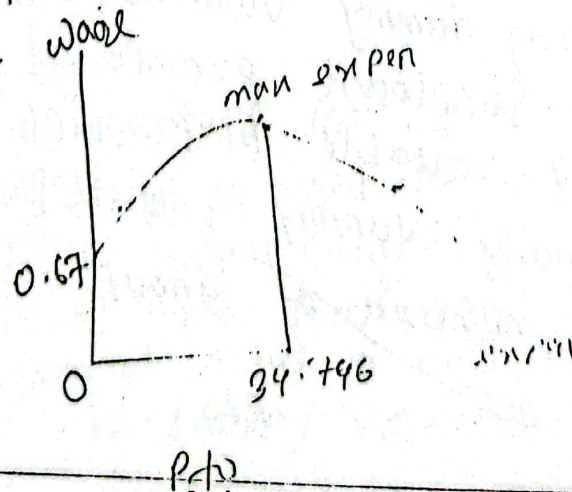
when educ = 20

$$\log(\hat{wage}) = 0.67 - 0.413(1) + 0.013(20) - 0.0056(20) \\ = 1.605$$

(i) The marginal effect of exper on wage will be negative at the following point.

$$\text{Here, } \text{exper} = \left| \frac{\hat{\beta}_9}{2\hat{\beta}_{10}} \right| \\ = \left| \frac{0.041}{2(0.00059)} \right| \\ = 34.746$$

It can be shown in the following figure -



after the 34.746 years exper wage rate tends to decline. At any point above $\text{exper} = 34.746$ marginal effect of experience on wage will be negative.

5) we know,

$$\begin{aligned}\text{adjusted } R^2(\bar{R}^2) &= 1 - (1 - R^2) \frac{n-1}{n-k} \\ &= 1 - (1 - .563) \frac{526-1}{526-5} \\ &= 0.56\end{aligned}$$

Interpretation:

Adjusted R^2 shows goodness of fit like R^2 . But it is adjusted with df for excluding the effect of irrelevant variable. Here for $k > 1$, $\bar{R}^2 < R^2$ means that as $\bar{R}^2$ is increasing on variable increasing but it increases less than that of R^2 .

The dummy variable trap is where we have a case of perfect collinearity in the sense that unless one dummy variable is omitted from the regression, the sum of all dummy variables add up to 1, giving and inaccurate answer.